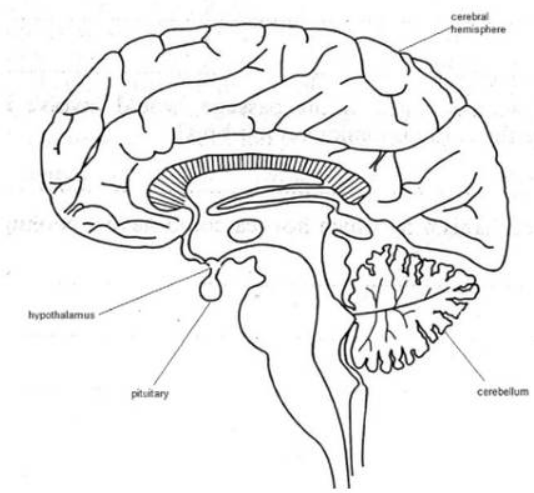
Type MS · Identify brain regions and match them to their functions

Answer 1 of 9 · Question 1 part b · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

number				
1	(a)	 correct labels ;;;;		4
	(b)	<table><tr><th>Function</th><th>Structure</th></tr><tr><td>control of muscle tone and posture</td><td>cerebellum</td></tr><tr><td>intelligence and thought</td><td>cerebral hemisphere</td></tr><tr><td>monitoring of body temperature</td><td>hypothalamus</td></tr><tr><td>release of ADH</td><td>Pituitary</td></tr></table>		4

Total 8 marks

Type MS · Identify brain regions and match them to their functions

Answer 1 of 9 · Question 1 part b · 4 marks

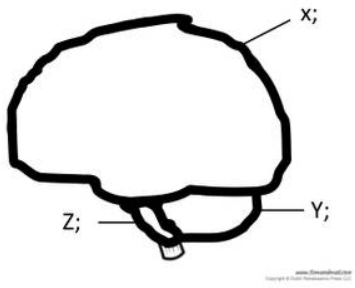
2021 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify brain regions and match them to their functions

Answer 2 of 9 · Question 7 part a · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

7 (a)		Letters can be anywhere within outline shown. Allow labels in place of than letters.	3
(b) (i)	Motor;		1
(ii)	Any one from the following: (sensory neurone): longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord (motor neurone) shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord	Answer must be comparative and state which type of neurone is being referred to	1
(iii)	• chemicals/neurotransmitters; • diffuse; • across a synapse; • bind to receptors (on post-synaptic membrane);		3
(c)	• stem cells are undifferentiated/unspecialised/can differentiate; • could be made to form nerve cells; • to replace damaged cells in brain;		1 1 1

Total question 7 = 11 marks

Type MS · Identify brain regions and match them to their functions

Answer 2 of 9 · Question 7 part a · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify brain regions and match them to their functions

Answer 3 of 9 · Question 6 part b · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

6(a)	short term is recent event/long term goes back many years; short term very little processing, long term has more processing;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(b)	cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum;		**(1)**
Question Number	Answer	Additional guidance	Mark
6(c)	• language/speaking/talking problems / repetitive actions/confusion; • loss of voluntary muscle function; • loss of intellectual capacity/function; • deterioration in social behaviour;		**(2 max)**
Question Number	Answer	Additional guidance	Mark
6(d)	• blocks synapse/cleft/build-up of plaques; • prevents diffusion of acetylcholine / transmitter/neurones communicating; • blocks post-synaptic receptors/prevents generation of new impulse; • death of neurones;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
6(e)	• inhibits/prevents breakdown of acetylcholine / transmitter; • increases chance of synaptic transmission / neurones communicating/generating new impulse; • reduces death of neurones;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(f)	formation of β-amyloid is caused by incorrect breakdown of APP; build-up/formation of β-amyloid (in neurones);		**(2)**

Total 12 Marks

Type MS · Identify brain regions and match them to their functions

Answer 3 of 9 · Question 6 part b · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Identify brain regions and match them to their functions

Answer 4 of 9 · Question 7 part a.i · 3 marks

2020 JAN · Paper 02 · find part a.i in this mark scheme extract

7 (a) (i)	<table><tr><th>Part</th><th>Function</th><th>Tick</th></tr><tr><td>medulla</td><td>controls heart rate and breathing</td><td>✓</td></tr><tr><td>pituitary gland</td><td>involved in language</td><td></td></tr><tr><td>hypothalamus</td><td>responsible for osmoregulation</td><td>✓</td></tr><tr><td>cerebellum</td><td>controls voluntary movements</td><td>✓</td></tr></table>	Maximum of two marks if all four boxes are ticked.	3
(ii)	• transmit impulses to effector organs/muscles/glands/controls movement/voluntary actions(1)		1
(b)	• lack of dopamine/no dopamine produced / released/release of dopamine inconsistent(1) • less nerve impulses generated/nerve impulses (to muscles/along motor neurones) inconsistent (1) • tremor/rigidity/slowness of movement/imbalance occur(1)	Allow neurotransmitter for dopamine	3

Total for Question 7 = 7 marks

Type MS · Identify brain regions and match them to their functions

Answer 5 of 9 · Question 7 part a.ii · 1 mark

2020 JAN · Paper 02 · find part a.ii in this mark scheme extract

7 (a) (i)	<table><tr><th>Part</th><th>Function</th><th>Tick</th></tr><tr><td>medulla</td><td>controls heart rate and breathing</td><td>✓</td></tr><tr><td>pituitary gland</td><td>involved in language</td><td></td></tr><tr><td>hypothalamus</td><td>responsible for osmoregulation</td><td>✓</td></tr><tr><td>cerebellum</td><td>controls voluntary movements</td><td>✓</td></tr></table>	Maximum of two marks if all four boxes are ticked.	3
(ii)	• transmit impulses to effector organs/muscles/glands/controls movement/voluntary actions(1)		1
(b)	• lack of dopamine/no dopamine produced / released/release of dopamine inconsistent(1) • less nerve impulses generated/nerve impulses (to muscles/along motor neurones) inconsistent (1) • tremor/rigidity/slowness of movement/imbalance occur(1)	Allow neurotransmitter for dopamine	3

Total for Question 7 = 7 marks

Type MS · Identify brain regions and match them to their functions

Answer 6 of 9 · Question 6 part a.i · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

6	(a)	(i)	A; (brain)		1
		(ii)	B; (endocrine)		1
	(b)	(i)	• 75 (from weight of man) (1) • 320 × 2 (extra from drinking beer) (1) • 715cm^3 (1)		3
		(ii)	• 715 - 75 (1) • (640 ÷ 75) × 100 (1) • = 853% (1)		3
	(c)		any two from • increases permeability of collecting duct (1) • more water reabsorbed (1) • into blood (1)		2
	(d)		any two from • red blood cells loses water (1) • become distorted (1) • less oxygen transported (1) • volume of plasma decreases (1) • blood becomes thicker/more difficult to circulate(1) • increase in blood pressure (1)	Ignore less water in blood Allow water potential decreases	2
	(e)		any three from • waste products continue to be produced (1) • toxic (1) • need to be removed in urine (1) • less reabsorption of water (1) • less ADH produced (1)	Allow named waste product(s)	3

Total 15 marks

Type MS · Identify brain regions and match them to their functions

Answer 6 of 9 · Question 6 part a.i · 1 mark

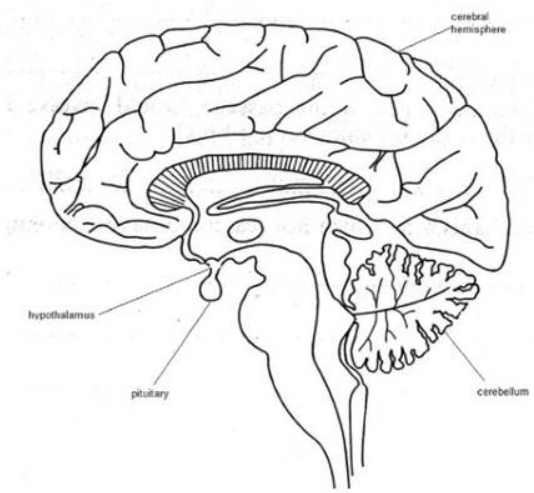
2020 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks
		Allow reverse for whole	

Type MS · Identify brain regions and match them to their functions

Answer 7 of 9 · Question 1 part a · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

number				
1	(a)	 correct labels ;;;;		4
	(b)	<table><tr><th>Function</th><th>Structure</th></tr><tr><td>control of muscle tone and posture</td><td>cerebellum</td></tr><tr><td>intelligence and thought</td><td>cerebral hemisphere</td></tr><tr><td>monitoring of body temperature</td><td>hypothalamus</td></tr><tr><td>release of ADH</td><td>Pituitary</td></tr></table>		4

Total 8 marks

Type MS · Identify brain regions and match them to their functions

Answer 7 of 9 · Question 1 part a · 4 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify brain regions and match them to their functions

Answer 8 of 9 · Question 3 part a · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

number			
3 (a)	C hypothalamus; A does not control sweating B does not control sweating D does not control sweating		1
(b)	B motor; A transmits impulse from sensory to motor C alternative name for A with same function D transmits impulse to brain		1
(c)	• (sweat) evaporates from (the surface of) the skin; • using heat from body/blood; • cools body/reduces body temperature;		3
(d)	• runs in families; • must be faulty/mutated gene/chromosome/allele; • which transfers/passes on information to offspring;		3
(e)	obesity / high sugar intake / high BMI;		1
(f)	• exercise/activity of individual varies; • greater rate of aerobic respiration; • heat given off as by-product/increased body temperature; • external temperature varies; • hotter more sweating/cooler less sweating; • mass of individual varies; • volume of water in the body/fluid intake varies;	Allow reverse argument for mps 2 and 3	5 Total 14

Type MS · Identify brain regions and match them to their functions

Answer 8 of 9 · Question 3 part a · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify brain regions and match them to their functions

Answer 9 of 9 · Question 3 part d · 1 mark

2024 JUN · Paper 02 · find part d in this mark scheme extract

Number			
3 (a)	Smoking cigarettes; Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ; OR lack of exercise; obesity/heart disease/high blood pressure/ type ii diabetes/eq;	Allow drugs or named drug and their effect on health for 2 marks.	(2)
(b)	Any two from: • production of bile reduced/affected; • less fat emulsified/fat not emulsified; • lipase less active/lipase unable to digest as much (fat)/slower lipase activity;	Allow bile not produced Allow description of emulsification	(2)
(c)	blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina;		(1)
(d)	cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata);	Allow any part of the brain that is affected by alcohol	(1)
(e)	Any three from: • reduces release of neurotransmitters; • reduced diffusion across synapse; • less neurotransmitters bind to receptors on post-synaptic membrane; • slows/ fewer transmission of impulses; • slower reaction /reflexes;		(3)
(f)	Any two from • increased heart rate; • increased breathing rate; • pupil dilation; • increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose; • increase in blood flow to brain/muscles; • increase in (aerobic) respiration (in muscles);		(2)

Total for Question 3 = 11 marks